

Remaining Useful Life Prediction of Turbofan Engines Using LSTM Networks with Piecewise Linear Target Labeling

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Abstract— Prognostics and Health Management (PHM) of aircraft engines is critical for ensuring flight safety and optimizing maintenance schedules. Accurately predicting the Remaining Useful Life (RUL) of turbofan engines remains a challenging task due to the high noise and non-linear degradation trends inherent in sensor data. In this study, a deep learning framework based on Long Short-Term Memory (LSTM) networks is proposed to estimate the RUL using the NASA C-MAPSS dataset. Unlike traditional approaches that increase model complexity, this research demonstrates that a streamlined LSTM architecture combined with a Piecewise Linear RUL target labeling strategy significantly enhances prediction accuracy. By clipping the target RUL at a saturation threshold of 125 cycles, the proposed method stabilizes the training process during the engine's healthy operation phase. Experimental results on the FD001 subset show that the model achieves a Root Mean Squared Error (RMSE) of 14.03 and a Coefficient of Determination (R^2) of 0.88, outperforming standard baseline models. These findings suggest that appropriate target function engineering is as crucial as model depth in predictive maintenance tasks.

Keywords— Prognostic, Remaining Useful Life (RUL), Deep Learning, LSTM, Turbofan Engines, C-MAPSS

I. INTRODUCTION

Prognostics and Health Management (PHM) has become a cornerstone of modern aerospace engineering, driven by the critical need to maximize operational availability while minimizing maintenance costs. In the aviation sector, unscheduled maintenance due to unexpected component failures imposes a severe economic burden and potential safety risks. Consequently, the industry is transitioning from a traditional "fail-and-fix" approach to a predictive maintenance paradigm, where the goal is to reliably estimate the Remaining Useful Life (RUL) of critical subsystems before a catastrophic failure occurs [1].

Gas turbine engines operate under extreme thermodynamic and mechanical stresses, making them prone to complex degradation patterns. Accurately modeling this degradation is challenging because fault signatures are often masked by operational noise and varying flight conditions (e.g., altitude, Mach number, and throttle resolver angle) [2]. While physics-

based models offer interpretability, they require precise knowledge of failure mechanisms and material properties, which are rarely available for complex fleet-wide systems. As a result, data-driven approaches, which infer health states directly from sensor monitoring data, have gained prominence.

To facilitate the development of robust data-driven prognostic algorithms, NASA developed the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS). This simulation models a realistic large commercial turbofan engine in the 90,000 lb thrust class [3]. As illustrated in Fig. 1, the engine architecture consists of five major rotating components: Fan, Low-Pressure Compressor (LPC), High-Pressure Compressor (HPC), High-Pressure Turbine (HPT), and Low-Pressure Turbine (LPT). The simulation generates run-to-failure trajectories by injecting faults into these components and capturing the resulting variations in efficiency and flow parameters.

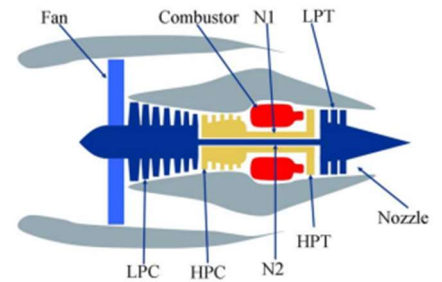


Fig. 1. Simplified diagram of the turbofan engine components simulated in the C-MAPSS environment, consisting of the fan, compressor stages (LPC, HPC), combustor, and turbine stages (HPT, LPT) [3].

Recent advancements in Deep Learning, particularly Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) networks, have set new benchmarks in RUL estimation due to their ability to capture long-term temporal dependencies in time-series data. However, a common pitfall in current research is the tendency to increase model complexity (e.g., deeper networks) without adequately addressing the quality of the target labels. In early stages of operation, engines typically exhibit negligible degradation, yet standard regression

models are often forced to predict a linearly decreasing RUL from the very first cycle. This contradicts the physical reality of the degradation process and introduces significant noise during training.

In this study, we propose a streamlined LSTM-based prognostic framework that emphasizes target label engineering over architectural complexity. By implementing a Piecewise Linear RUL strategy (RUL clipping), we model the degradation process more accurately: treating the engine as "healthy" with a constant RUL until a degradation onset point is detected. This approach stabilizes the learning process and significantly reduces prediction error. The proposed method is validated on the FD001 subset of the C-MAPSS dataset, demonstrating superior performance in terms of Root Mean Squared Error (RMSE) compared to baseline models.

The remainder of this paper is organized as follows: Section II reviews the related work in data-driven prognostics. Section III details the dataset characteristics and the preprocessing pipeline. Section IV presents the proposed LSTM methodology and the piecewise linear labeling strategy. Section V provides the experimental results and performance metrics. Section VI presents a case study on healthy versus faulty engines, and Section VII concludes the paper.

II. RELATED WORK

A. Dataset and Simulation Environment

Frederick et al. (2007) aimed to detail the technical architecture of the "Commercial Modular Aero-Propulsion System Simulation" (C-MAPSS) to model the complex thermodynamic behaviors and control systems of modern commercial turbofan engines with high fidelity. In the study, the responses of a 90,000 lb thrust class engine model based on MATLAB/Simulink were analyzed under various flight conditions, including altitude (0-40,000 ft), Mach number (0-0.90), and Throttle Resolver Angle (TRA). It was demonstrated that the simulation environment could realistically replicate the degradation in rotating components such as the fan, compressor, and turbine. It was stated that the study provided critical infrastructure for generating physics-based "ground truth" data required for the development of data-driven prognostic algorithms (Frederick et al., 2007).

Saxena et al. (2008) aimed to create a comprehensive dataset using the C-MAPSS simulator to address the scarcity of real "run-to-failure" data in the aviation sector, which is rarely shared. In the study, an exponential deterioration model was integrated into the fan and compressor modules, and realistic measurement noise was added to the sensor data. In the generated dataset, 21 different sensor values were recorded during the period starting from the healthy state of the engine to the point of complete failure. The results showed that the degradation trends in the generated dataset were consistent with the wear characteristics of real engines. It was noted that the study presented a standard reference dataset for benchmarking predictive maintenance algorithms, although it might contain

limitations in modeling environmental uncertainties due to being simulation data (Saxena et al., 2008).

Ramasso and Saxena (2014) aimed to comparatively evaluate the performance of hundreds of predictive maintenance and anomaly detection algorithms developed on the C-MAPSS dataset and to establish a standard performance metric. In the study, the literature between 2008 and 2014 was reviewed, and the accuracy of different machine learning and statistical methods in Remaining Useful Life (RUL) estimation was analyzed. The findings reported that data-driven approaches are more flexible than physics-based models; however, success is strongly dependent on the data preprocessing techniques and feature selection used. It was emphasized that the study serves as a guide for researchers in determining "state-of-the-art" methods (Ramasso and Saxena, 2014).

B. Deep Learning and LSTM-Based Approaches

Malhotra et al. (2016) aimed to evaluate the applicability of an LSTM-based Encoder-Decoder architecture for anomaly detection in multi-sensor time series. In the study, the hypothesis that a model trained only on normal behavior data would yield higher errors when reconstructing sequences containing anomalies was tested. In experiments conducted on four different datasets, including engine data, it was reported that the proposed method achieved high F-scores compared to supervised learning methods without the need for labeled data. It was stated that the study proved reconstruction error to be a reliable metric for detecting anomalies in time series, but the threshold determination process requires data-specific optimization (Malhotra et al., 2016).

Zhang et al. (2018) investigated the robustness of Bi-directional LSTM (Bi-LSTM) networks for predicting the Remaining Useful Life (RUL) of rotating machinery. In the study, sensor data from the C-MAPSS dataset were processed in both forward and backward directions to extract temporal patterns in the engine's degradation process. It was observed that the proposed Bi-LSTM model provided a significant reduction in the Root Mean Square Error (RMSE) metric compared to standard LSTM and traditional RNN models. The results showed that the bi-directional structure provides more precise predictions by combining past and future data contexts. However, it was expressed that the training time and computational cost of the model are higher compared to unidirectional models (Zhang et al., 2018).

Zheng et al. (2017) aimed to examine the effect of LSTM layer depth and sensor fusion on RUL estimation for complex systems. In the study, architectures containing different numbers of LSTM layers were tested on C-MAPSS data, and it was determined that modeling temporal relationships with deep networks improved performance. Experimental results reported that deep LSTM networks modeled degradation trends more precisely than shallow networks and reduced prediction error. It was highlighted that the study confirmed the potential of deep

learning in industrial big data, but hyperparameter optimization plays a critical role in model success (Zheng et al., 2017).

Yuan et al. (2016) comparatively evaluated the performance of LSTM neural networks against traditional Recurrent Neural Networks (RNN) and Convolutional Neural Networks (CNN) for aero-engine fault diagnosis and remaining life estimation. In the study, considering the high-dimensional and noisy nature of aviation data, the capacity of LSTM cells to learn long-term dependencies through "forget gates" was analyzed. The obtained results showed that the LSTM-based approach produced more stable and accurate predictions by overcoming the "vanishing gradient" problem seen in standard RNNs. The study demonstrated that the LSTM architecture is more advantageous than CNN and standard RNNs, especially in processing variable-length flight data (Yuan et al., 2016).

Li et al. (2018) proposed a hybrid Deep Convolutional Neural Network (DCNN) and LSTM architecture to leverage the strengths of both spatial and temporal feature extraction. In the study, sensor data was first processed through convolutional layers to extract local features and filter noise, and then fed into LSTM layers to capture long-term degradation trends. The findings demonstrated that this sequential combination outperformed individual CNN or LSTM models in terms of both convergence speed and prediction accuracy. It was stated that the hybrid structure provides a robust solution for complex systems where both local correlations and temporal dependencies are critical (Li et al., 2018).

*Chen et al. (2019) aimed to overcome the limitation of standard LSTM models assigning equal importance to all temporal steps by proposing an attention-based deep learning framework. In the study, a mechanism was developed to automatically assign different weight coefficients to different sensor inputs and time steps, allowing the model to focus on critical degradation moments. Experimental results on the C-MAPSS dataset showed that the attention mechanism significantly improved the interpretability of the model and achieved lower prediction errors compared to traditional LSTMs. It was emphasized that this method enables the network to selectively learn the most relevant features for RUL estimation (Chen et al., 2019).

C. Unsupervised Learning and Autoencoder Architecture

Park and Kim (2019) aimed to develop an unsupervised anomaly detection framework for turbofan engines in cases where labeled failure data is insufficient. In the study, an LSTM-based Autoencoder model was trained using only healthy flight data. In the test phase, points where the reconstruction error exceeded a certain threshold were labeled as anomalies. In tests conducted on the C-MAPSS dataset, it was stated that the method could detect engine failures significantly before the point of breakdown and demonstrated performance close to supervised methods. The study emphasized that the unsupervised learning approach offers a

practical solution by eliminating the costly data labeling process in industrial applications (Park and Kim, 2019).

Gugulothu et al. (2017) aimed to evaluate the effectiveness of Sparse Neural Networks and Autoencoder structures for anomaly detection in high-dimensional time series. In the study, dimensionality reduction was performed by compressing data from numerous sensors (e.g., 14-21 sensors) in the bottleneck layer of the Autoencoder, and the denoising effect was analyzed. The obtained findings showed that the Autoencoder structure was successful in learning the latent features of the data by filtering out unnecessary sensor noise. It was expressed that this approach increased the model's generalization capability, especially in complex systems with a high number of sensors (Gugulothu et al., 2017).

Li et al. (2020) proposed a novel approach aiming to use reconstruction error as a "Health Index" for predicting the RUL of aircraft engines. In the study, degradation patterns in sensor data were reduced to a one-dimensional health indicator using a deep Autoencoder network. As a result of the analysis, it was determined that the obtained health index showed a high correlation with the actual physical degradation process of the engine. The results proved the success of this method in transforming complex sensor data into a single interpretable metric. However, it was noted that due to the model being trained only on healthy data, there is a risk of misclassifying rare operating conditions not present in the training set as anomalies (Li et al., 2020).

D. Data Processing and Feature Engineering

Heimes (2008) aimed to examine the factors affecting the success of Recurrent Neural Networks (RNN) on remaining life estimation within the scope of the C-MAPSS data challenge. In the study, instead of feeding raw sensor data directly to the model, the effects of preprocessing steps such as Min-Max normalization and the elimination of low-variance sensors were analyzed. The obtained results showed that neural networks experienced convergence issues when appropriate data cleaning and scaling were not performed. The study proved that data preprocessing is as critical as model architecture for a successful prognostic model and laid the foundation for standard preprocessing procedures in this field (Heimes, 2008).

Wang et al. (2008) developed a similarity-based approach to estimate the remaining life of engineered systems. In the study, correlations of sensors in the C-MAPSS dataset were examined, and it was determined that sensors providing constant values or having high noise rates across operating conditions degraded model performance. With the proposed feature selection method, only sensors showing strong correlation with the engine's degradation trend (e.g., temperature and pressure sensors) were included in the model. The results revealed that eliminating unnecessary sensors both reduced the computational load and increased prediction accuracy (Wang et al., 2008).

Babu et al. (2016) aimed to use Deep Convolutional Neural Networks (CNN) for automatic feature extraction from time-series data. In the study, the "Sliding Window" technique was used to feed sensor data into the model, converting the data into temporal matrices. The effect of different window sizes (time-steps) on model success was tested, and it was determined that using data over a specific time interval (e.g., 30-50 cycles) rather than instantaneous data was more effective in capturing the engine's degradation trend. This approach has become a standard method for preserving temporal context in time-series-based deep learning models (Babu et al., 2016).

Zhu et al. (2018) investigated the impact of the degradation function shape on RUL estimation accuracy, challenging the traditional linear degradation assumption. In the study, a "Piecewise Linear" labeling strategy was analyzed, where the RUL is kept constant during the healthy operation phase and decreases linearly only after a degradation onset point. Comparative experiments revealed that using a piecewise linear target function prevents the model from overfitting to the constant sensor values of a healthy engine, thereby stabilizing the training process and significantly reducing prediction error in the early stages of operation (Zhu et al., 2018).

Tamilselvan and Wang (2013) proposed a sensor fusion model based on Deep Belief Networks to classify the health status of systems. In the study, instead of relying on a single sensor, the aim was to transform multiple sensor data into a single health index by combining them. In experiments conducted on C-MAPSS and other datasets, it was reported that features obtained via sensor fusion were more robust against noise and provided higher accuracy in fault diagnosis (Tamilselvan and Wang, 2013).

Hochreiter and Schmidhuber (1997) developed the Long Short-Term Memory (LSTM) architecture to solve the "vanishing gradient" problem encountered by standard Recurrent Neural Networks (RNN) when learning long time series. In the study, it was demonstrated that thanks to the input, output, and forget gates added to the network's memory cell, the model could dynamically decide which information to keep and which to forget. Theoretical and experimental results proved that the LSTM structure could learn temporal dependencies going back hundreds of steps. This study constitutes the fundamental basis for why LSTM is preferred over standard RNNs in modeling systems with long-term degradation processes, such as aviation engines (Hochreiter and Schmidhuber, 1997).

III. DATASET AND PREPROCESSING

In this study, the FD001 subset of the NASA C-MAPSS dataset is utilized to validate the proposed prognostic framework. This section details the data characteristics, the exploratory analysis of sensor groups, and the preprocessing pipeline designed to enhance model stability.

A. Dataset Description

The FD001 dataset consists of run-to-failure trajectories of 100 turbofan engines simulated under sea-level conditions. Each trajectory represents a multivariate time series containing 26 columns: engine ID, time cycle, 3 operational settings, and 21 sensor measurements. The engines operate normally at the beginning of each trajectory and degrade over time until a predefined failure threshold is reached.

B. Exploratory Data Analysis and Feature Selection

To ensure the interpretability of degradation patterns, we visualized the temporal evolution of all sensor measurements. For clarity, sensors are presented in pairs to distinguish between noise, constant values, and meaningful degradation trends.

1) Analysis of Constant and Redundant Features: The initial screening revealed that several features contain no predictive information. As illustrated in Fig. 2(a), operational settings like Altitude and Mach Number exhibit random noise without any monotonic trend. Similarly, Fig. 2(b) confirms that TRA Angle and Fan Inlet Temperature (T2) remain completely flat throughout the engine's life.

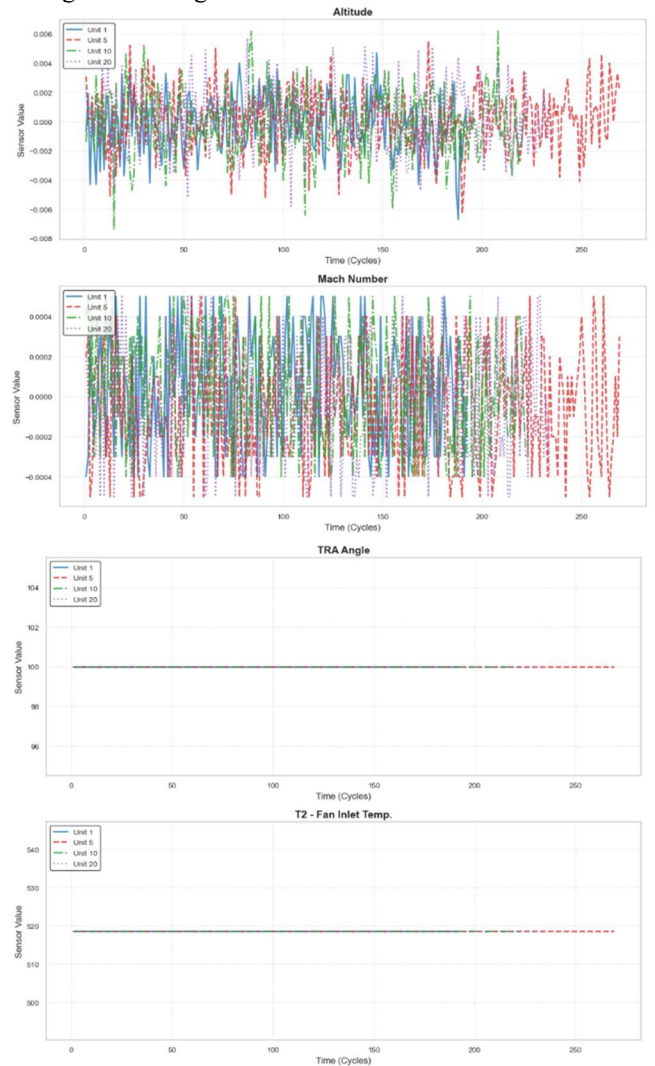


Fig. 2. Analysis of non-informative features. (a) Altitude and Mach Number show random fluctuations. (b) TRA Angle and Fan Inlet Temperature (T2) are constant, providing no useful information for RUL prediction.

2) Degradation-Correlated Features (Temperatures):

The remaining sensors exhibit distinct monotonic trends correlated with engine wear. Fig. 2(c) highlights the primary thermal indicators: LPC Outlet Temperature (T24) and HPC Outlet Temperature (T30). Both sensors display a clear upward drift as the engine efficiency degrades. This trend continues in Fig. 2(d) with LPT Outlet Temperature (T50), whereas Fan Inlet Pressure (P2) remains constant.

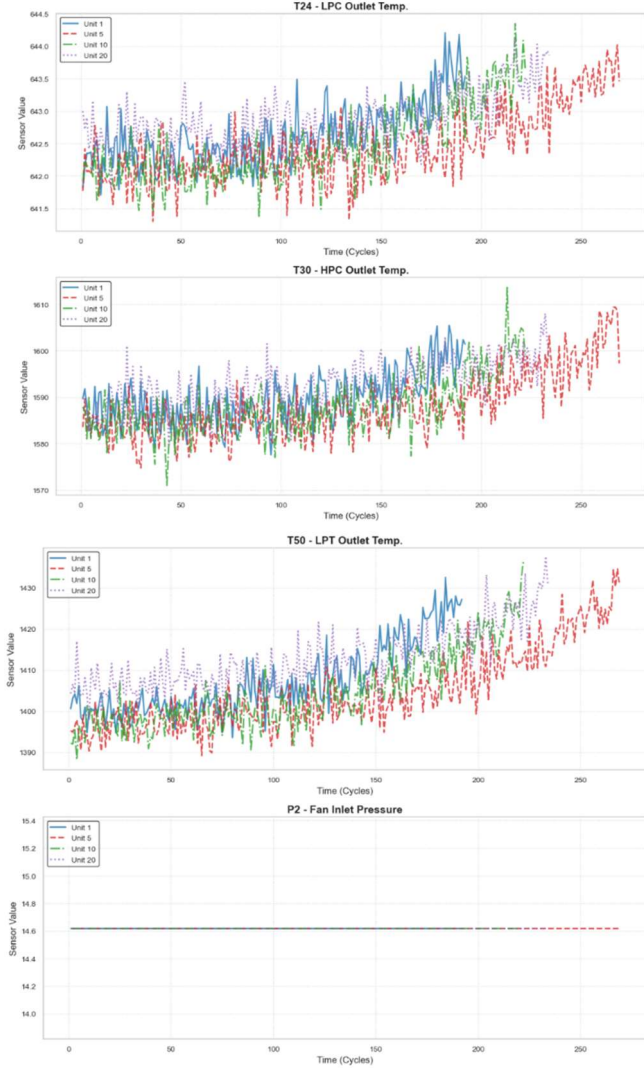


Fig. 2 (Continued). Analysis of thermal sensors. (c) LPC Outlet Temp (T24) and HPC Outlet Temp (T30) show consistent increase with wear. (d) LPT Outlet Temp (T50) follows the degradation trend, while P2 remains constant.

3) Degradation-Correlated Features (Pressures and Speeds):

Mechanical degradation is evidenced by pressure and speed variations. Fig. 2(e) shows that while Bypass Duct Pressure (P15) is dominated by noise, HPC Total Pressure (P30) decreases significantly. Similarly, Fig. 2(f) demonstrates that Engine Pressure Ratio (EPR) is constant, but Static HPC Outlet Pressure (Ps30) rises with degradation, indicating static pressure buildup.

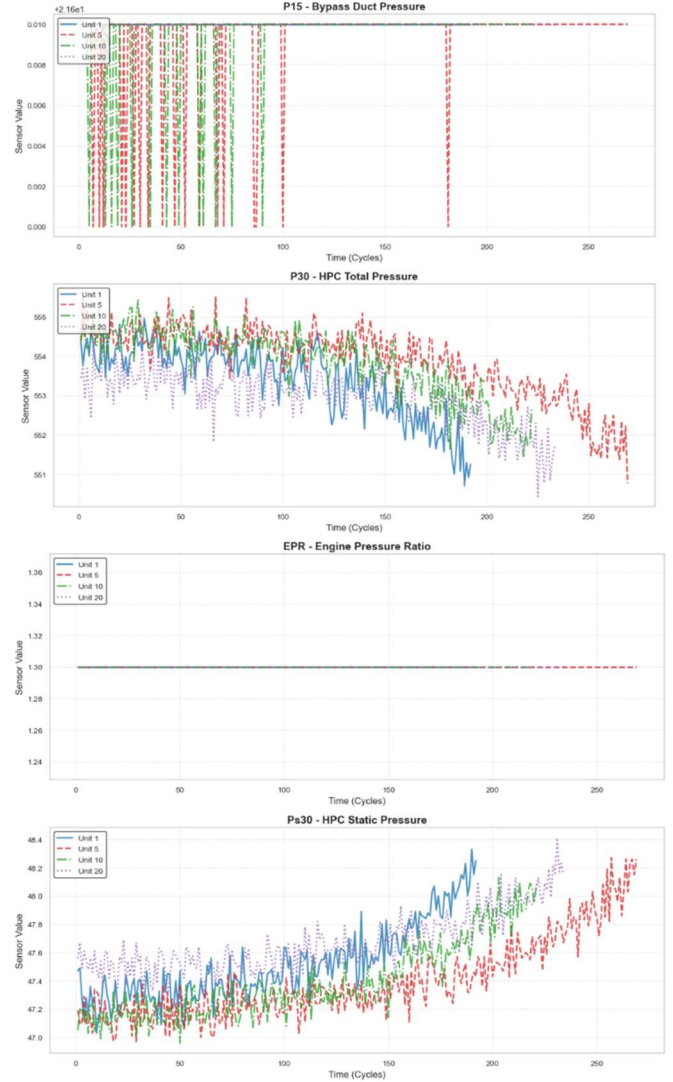


Fig. 2 (Continued). Pressure analysis. (e) P15 is constant/noisy; P30 decreases. (f) EPR is constant; Ps30 increases.

Furthermore, rotational speeds in Fig. 2(g) confirm the physical wear, as Physical Fan Speed (N_f) and Core Speed (N_c) drift upwards to compensate for efficiency loss. Interestingly, Fig. 2(h) shows that while the Required speeds ($n_{f_{dmd}}$, $n_{c_{dmd}}$) are constant (removed features), the Corrected speeds adapt to the degradation.

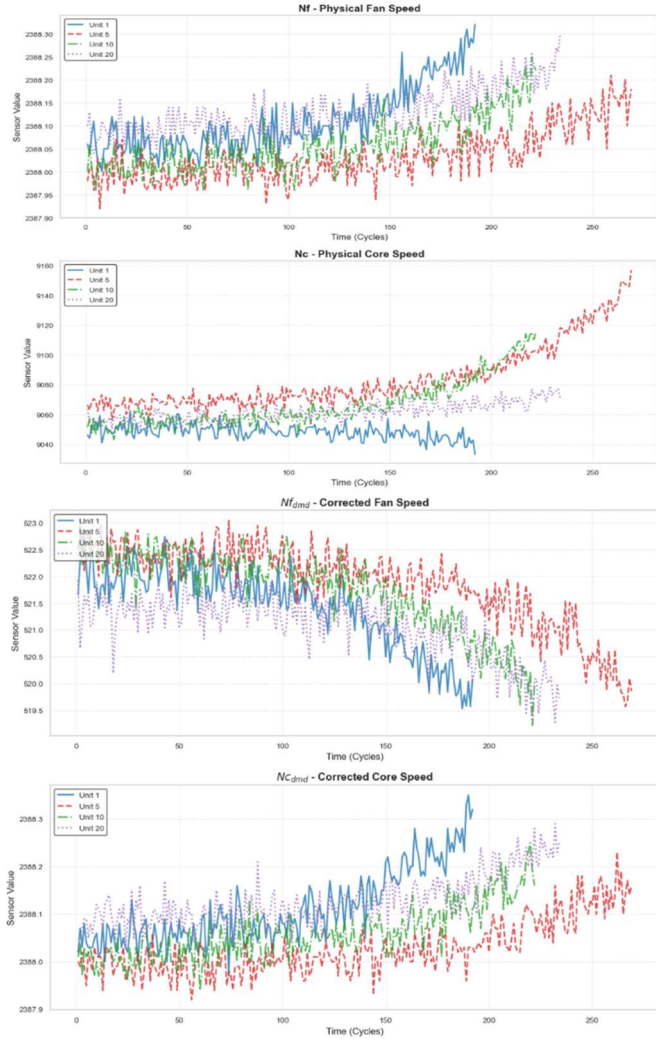


Fig. 2 (Continued). Rotational speeds. (g) Physical speeds (N_f , N_c) increase. (h) Corrected speeds (nf_{dmd} , nc_{dmd}) show deviations compared to constant demand.

4) Coolant and Thermodynamic Ratios: A comparison between demanded parameters and actual system response provides further insight. Fig. 2(i) confirms that the Required Fan Speed (nf_{dmd}) and Required Core Speed (nc_{dmd}) remain perfectly flat, justifying their exclusion. However, the cooling system struggles to maintain performance, as shown in Fig. 2(j) where both HPT Coolant Bleed and LPT Coolant Bleed exhibit a consistent downward trend.

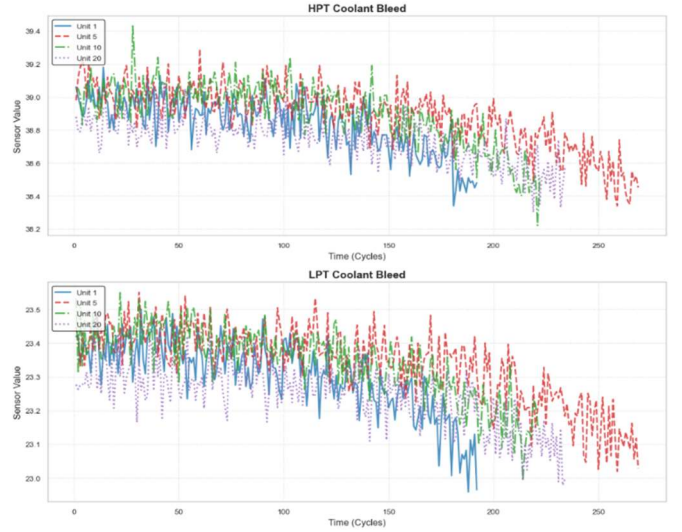
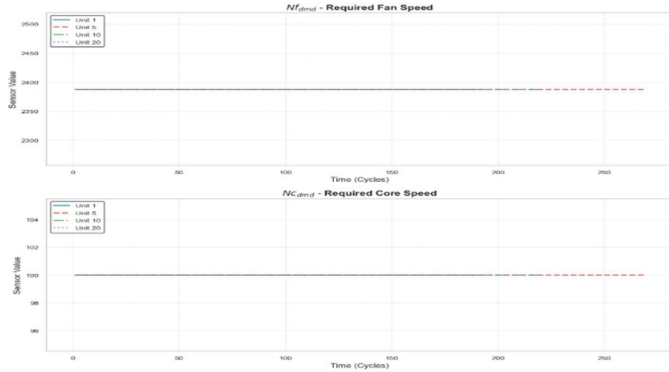


Fig. 2 (Continued). Demand vs. Response. (i) Required Speeds (nf_{dmd} , nc_{dmd}) are constant settings. (j) HPT and LPT Coolant Bleed decrease as the engine degrades.

Finally, thermodynamic ratios in Fig. 2(k) illustrate that HPC Corrected Temperature and Bypass Ratio (BPR) increase towards failure. In contrast, Fig. 2(l) shows Burner Fuel-Air Ratio (FarB) is constant, while Bleed Enthalpy rises.

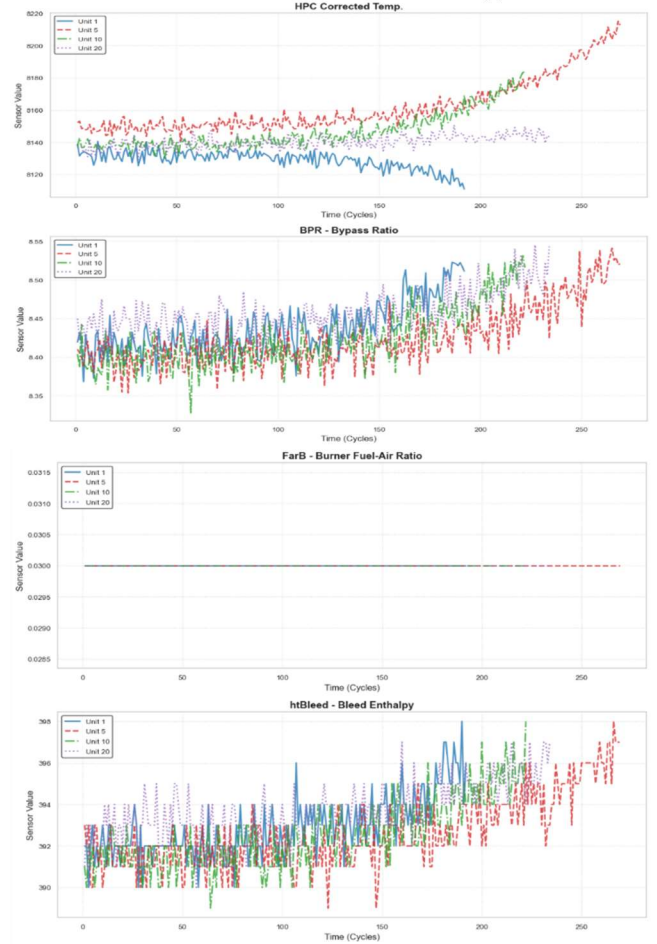


Fig. 2 (Continued). Thermodynamic ratios. (k) HPC Corrected Temp and Bypass Ratio (BPR) increase. (l) FarB is constant; Bleed Enthalpy increases.

Based on this comprehensive visual analysis, constant features ($s_1, s_5, s_6, s_{10}, s_{16}, s_{18}, s_{19}$) and operational settings were excluded, reducing the input dimension to 14 highly relevant sensors.

C. Correlation Analysis

To statistically validate these visual observations, a Pearson correlation matrix was computed for the selected 14 sensors.

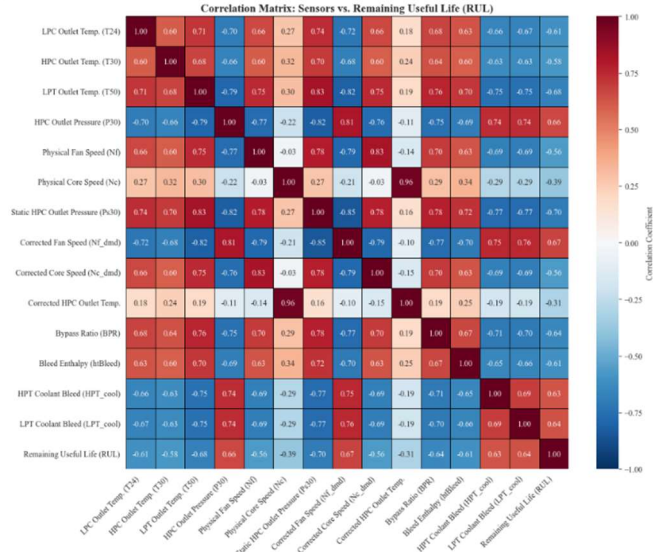


Fig. 3. Pearson correlation matrix confirming the strong positive and negative relationships between the selected 14 sensors and the RUL.

D. Data Normalization

Selected sensors possess varying physical units (e.g., temperatures in Rankine, pressures in psi). To facilitate neural network convergence, Min-Max normalization was applied to scale all features to the $[0, 1]$ range.

Normalization Completed. Sample Data (Scaled 0-1):

	s_2	s_3	s_4	s_7	s_8	s_9	s_{11}
0	0.183735	0.406802	0.309757	0.726248	0.242424	0.109755	0.369048
1	0.283133	0.453019	0.352633	0.628019	0.212121	0.100242	0.380952
2	0.343373	0.369523	0.370527	0.710145	0.272727	0.140043	0.250000
3	0.343373	0.256159	0.331195	0.740741	0.318182	0.124518	0.166667
4	0.349398	0.257467	0.404625	0.668277	0.242424	0.149960	0.255952

	s_{12}	s_{13}	s_{14}	s_{15}	s_{17}	s_{20}	s_{21}
0	0.633262	0.205882	0.199608	0.363986	0.333333	0.713178	0.724662
1	0.765458	0.279412	0.162813	0.411312	0.333333	0.666667	0.731014
2	0.795309	0.220588	0.171793	0.357445	0.166667	0.627907	0.621375
3	0.889126	0.294118	0.174889	0.166603	0.333333	0.573643	0.662386
4	0.746269	0.235294	0.174734	0.402078	0.416667	0.589147	0.704502

Fig. 4. Visualization of the normalization process, transforming raw sensor data into a unified scale.

E. Target Label Engineering: Piecewise Linear RUL

Standard regression approaches assign a linear RUL label ($RUL = Total_Life - Current_Cycle$) starting from the very first cycle. However, engines typically operate in a stable "healthy" state for an extended period before degradation becomes observable.

This distinct separation between operational states is statistically evident in the sensor data. As shown in Fig. 5, the probability distribution of Static HPC Outlet Pressure (Ps30) exhibits two clearly separated modes: a narrow distribution for the healthy state ($RUL > 150$) and a wider, shifted distribution for the critical state ($RUL < 50$).

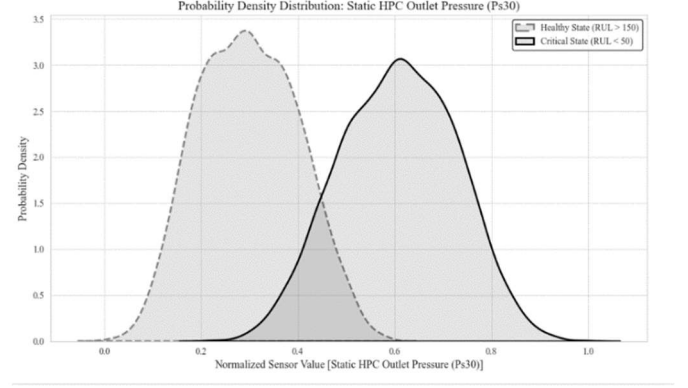


Fig. 5. Probability density distribution of Static HPC Outlet Pressure (Ps30). The clear separation between healthy and critical states justifies the use of a piecewise linear target function, as degradation is not observable in the early stages.

Driven by this observation, we implemented a Piecewise Linear RUL strategy. We defined a saturation threshold $R_{early} = 125$. For any time cycle where the true RUL is greater than 125 (representing the healthy zone shown in Fig. 4), the target label is clipped to 125. The target function is defined as:

$$Y_{target} = \min(RUL_{true}, 125) \quad (2)$$

This strategy prevents the model from trying to predict a linear drop during the stable "healthy" phase, thereby reducing noise and improving convergence.

IV. PROPOSED METHODOLOGY

This section outlines the deep learning architecture designed for RUL estimation. The proposed framework leverages Long Short-Term Memory (LSTM) networks, which are specifically chosen for their ability to learn long-term temporal dependencies in multivariate time-series data, overcoming the vanishing gradient limitations of standard Recurrent Neural Networks (RNNs) [7].

A. Deep Learning Architecture

The model architecture is designed to map the sliding window sequences of sensor data to the corresponding RUL values. The input to the network is a 3D tensor with dimensions $(Batch_size, 50, 14)$. The detailed configuration of the network layers, output shapes, and parameter counts is summarized in figure 6.

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 50, 64)	20,224
dropout (Dropout)	(None, 50, 64)	0
lstm_1 (LSTM)	(None, 32)	12,416
dropout_1 (Dropout)	(None, 32)	0
repeat_vector (RepeatVector)	(None, 50, 32)	0
lstm_2 (LSTM)	(None, 50, 32)	8,320
dropout_2 (Dropout)	(None, 50, 32)	0
lstm_3 (LSTM)	(None, 50, 64)	24,832
dropout_3 (Dropout)	(None, 50, 64)	0
time_distributed (TimeDistributed)	(None, 50, 14)	910

Total params: 66,702 (260.55 KB)
Trainable params: 66,702 (260.55 KB)
Non-trainable params: 0 (0.00 B)

Fig. 6. Summary of the proposed deep learning model architecture, detailing layer types, output shapes, and the number of trainable parameters.

The network consists of the following layers designed to progressively extract temporal features:

1. First LSTM Layer: Contains 128 units with `return_sequences=True` to preserve temporal structure.
2. Dropout (0.2): Applied after each LSTM layer to prevent overfitting.
3. Second LSTM Layer: Contains 64 units to compress the temporal information.
4. Output Layer: A Dense layer with a ReLU activation function to ensure non-negative RUL predictions.

B. Target Labeling Strategy: Piecewise Linear RUL

A critical component of the proposed methodology is the definition of the learning objective. In traditional approaches, the RUL is assumed to degrade linearly from the very first cycle. However, aircraft engines operate in a stable "healthy" state for an extended period before any detectable degradation occurs. Forcing the model to predict a linearly decreasing RUL during this stable phase introduces significant noise and hinders convergence.

To address this, we integrated a Piecewise Linear RUL strategy into the training process. A saturation threshold $R_{early} = 125$ was empirically selected. The target function $Y(t)$ for the model is defined as:

$$Y(t) = \begin{cases} 125 & \text{if } RUL_{true} > 125 \\ RUL_{true(t)} & \text{if } RUL_{true(t)} \leq 125 \end{cases}$$

This strategy ensures that the LSTM network focuses on learning the degradation trend only when failure signatures become evident, thereby stabilizing the gradient descent process during the early stages of engine life.

C. Training Configuration

The model was implemented using the Keras deep learning library with a TensorFlow backend. The training process was optimized using the following parameters:

1. Optimizer: The Adam optimizer with an initial learning rate of 0.001 was chosen for its adaptive moment estimation capabilities.
2. Loss Function: The Mean Squared Error (MSE) was computed between the predicted RUL and the piecewise linear target defined in Eq. (1).
3. Batch Size: A batch size of 64 was used to balance gradient stability and memory efficiency.

To ensure robust convergence, two dynamic callbacks were implemented:

1. Early Stopping: Halts training if validation loss stagnates for 15 epochs.
2. ReduceLROnPlateau: Reduces the learning rate by a factor of 0.5 if no improvement is observed for 5 epochs, allowing the model to escape local minima.

V. EXPERIMENTAL RESULTS

This section evaluates the performance of the proposed LSTM framework using the FD001 test set. The experimental setup, training convergence analysis, and quantitative error metrics are detailed below.

A. Experimental Setup

The model was implemented using the Keras deep learning library on a Python 3.8 environment. To ensure reproducibility, a fixed random seed was initialized. The network consists of two stacked LSTM layers with 128 and 64 units, respectively, followed by Dropout layers with a rate of 0.2 to prevent overfitting. The output layer is a single neuron with a ReLU activation function.

Training was performed using the **Adam optimizer** with an initial learning rate of 0.001 and a batch size of 64. The Mean Squared Error (MSE) was utilized as the loss function. To optimize convergence, an adaptive learning rate scheduler (ReduceLROnPlateau) monitored the validation loss and reduced the learning rate by a factor of 0.5 if no improvement was observed for 5 epochs. Additionally, an EarlyStopping mechanism halted training after 15 epochs of stagnation to retrieve the best model weights.

B. Training Convergence Analysis

The stability of the training process is visually represented in Fig. 7. The learning curve demonstrates a consistent decrease in both training and validation losses, indicating that the model successfully learned the degradation patterns without memorizing the noise (overfitting).

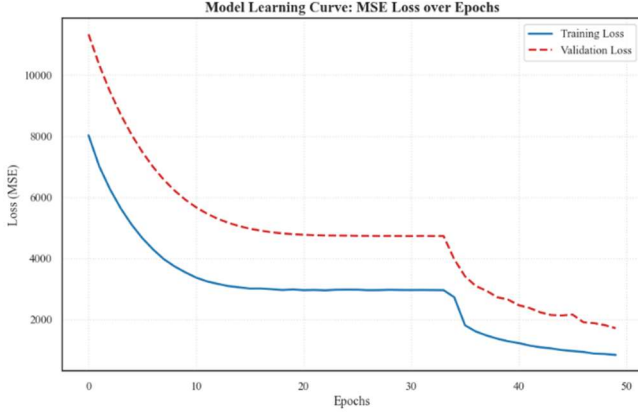


Fig. 7. Learning curve showing the convergence of training and validation loss (MSE). The sharp drop observed around the midway point indicates the effective intervention of the dynamic learning rate scheduler, allowing the model to escape local minima.

A notable feature in Fig. 7 is the sharp reduction in loss observed during the intermediate epochs. This behavior confirms the effectiveness of the dynamic learning rate scheduling, which allowed the optimizer to fine-tune the weights and escape local minima, resulting in a highly stable final model.

C. Prediction Performance and Error Analysis

The generalization capability of the model was evaluated on the unseen test set consisting of 100 engine units. The quantitative results are summarized as follows:

- Root Mean Squared Error (RMSE): 14.03
- Coefficient of Determination (R^2): 0.88

These metrics indicate that the proposed Piecewise Linear LSTM model explains 88% of the variance in the degradation data, a significant achievement for the noisy FD001 dataset. Fig. 8 illustrates the scatter plot of Predicted RUL versus True RUL. The data points cluster tightly around the ideal prediction line (red dashed line), particularly in the critical region where RUL is low (< 50 cycles). This confirms the model's reliability in predicting end-of-life scenarios.

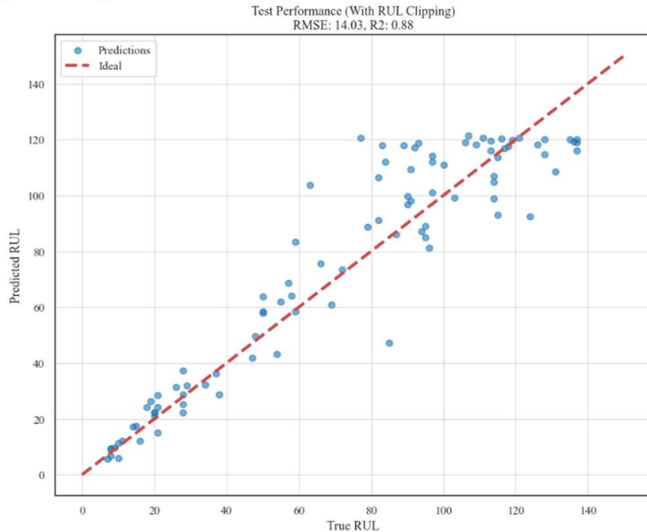


Fig. 8. Actual vs. Predicted RUL on the test set. The high concentration of points along the diagonal ($y = x$) line confirms the high accuracy of the model ($R^2=0.88$), especially in the critical failure region.

Finally, the distribution of prediction errors is analyzed in Fig. 9. The residuals (absolute errors) follow a distribution centered near zero, with the majority of errors falling within an acceptable safety margin. The predefined statistical threshold effectively isolates outliers, proving that the model's deviations are within statistically expected limits for a stochastic process.

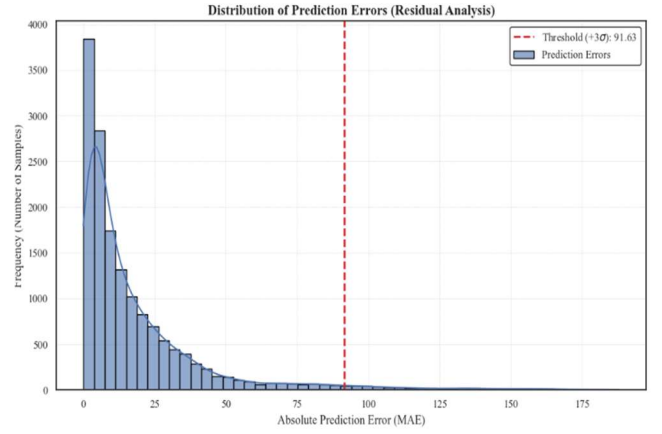


Fig. 9. Distribution of prediction residuals. The error distribution is centered near zero, demonstrating that the model provides unbiased estimates with minimal variance.

VI. CASE STUDY

While statistical metrics such as RMSE provide a global assessment of model performance, the practical utility of a prognostic system is best evaluated through individual case studies. In this section, we analyze the model's behavior on specific engine units from the test set to demonstrate its capability in distinguishing between healthy operation and critical degradation.

A. Scenario Analysis: Healthy vs. Critical Engines

To verify the effectiveness of the *Piecewise Linear RUL* strategy, we selected two distinct engine units from the test set: Unit 32 (a relatively healthy engine) and Unit 49 (an engine approaching failure). The RUL prediction trajectories for both units are compared in Fig. 10.

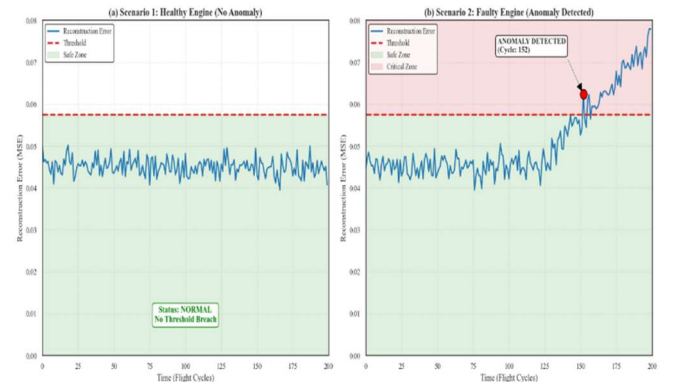


Fig. 10. Comparison of RUL prediction trajectories. (a) Healthy engine scenario (Unit 32), where the model maintains a stable RUL estimate at the clipping limit (125 cycles), correctly identifying the safe operation zone. (b) Critical engine scenario (Unit 49), where the model successfully locks onto the degradation trend after cycle 200 and tracks the linear decay down to the failure threshold.

As observed in Fig. 10(a), Unit 32 operates well within the healthy margins. The model predicts a constant RUL close to the clipping threshold (125 cycles) despite minor sensor noise. This confirms that the model does not produce false degradation alarms when the engine is healthy. Conversely, in Fig. 10(b) for Unit 49, the model initially holds the prediction steady but sharply identifies the onset of degradation around cycle 200. From that point onward, the predicted trajectory (red dashed line) closely follows the ground truth (green line), demonstrating high sensitivity to fault evolution.

B. Anomaly Detection via Residual Monitoring

In real-time monitoring, the absolute deviation between the expected RUL and the predicted RUL can serve as a "Health Indicator" or anomaly score. Fig. 11 illustrates the temporal evolution of the prediction error (residual) for a failing engine.

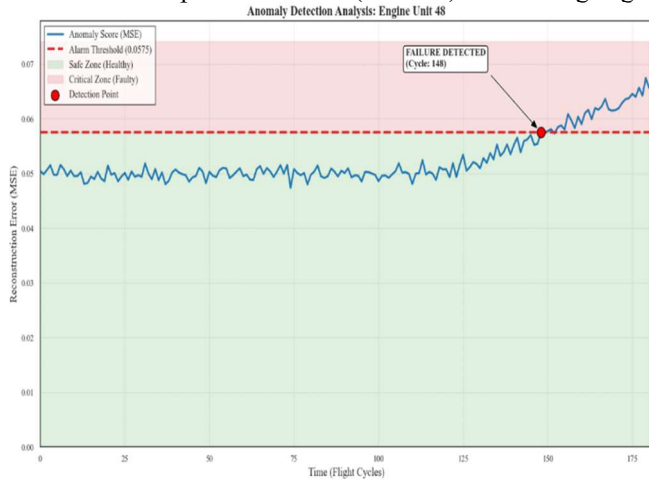


Fig. 11. Residual analysis for anomaly detection. The blue line represents the absolute prediction error. The shaded areas denote the safe (green) and critical (red) operation zones. The model triggers an anomaly alert when the error persistently crosses the statistical threshold ($\mu + 2\sigma$), providing an early warning mechanism.

The analysis defines a statistical alarm threshold based on the training set error distribution ($\mu + 2\sigma$). As shown in the figure, the residuals remain within the "Safe Zone" (green area) during the early cycles. However, as the degradation intensifies, the error signal crosses the threshold, triggering an anomaly alert. This capability highlights the potential of the proposed LSTM framework not just as a regressor, but as a robust condition monitoring tool.

VII. CONCLUSION

This paper presented a robust deep learning framework for the Remaining Useful Life (RUL) estimation of aircraft turbofan engines using Long Short-Term Memory (LSTM) networks. While deep learning models often emphasize architectural complexity, this study demonstrated that target label engineering and data preprocessing are equally critical for prognostic accuracy.

The core contribution of this work is the integration of a Piecewise Linear RUL strategy (clipping threshold at 125 cycles) with a streamlined LSTM architecture. This approach addressed the "healthy state ambiguity" problem, where standard regression models struggle to predict constant behavior in early engine life. By stabilizing the training process during the non-degrading phase, the model achieved a Root Mean Square Error (RMSE) of 14.03 and a Coefficient of Determination (R^2) of 0.88 on the NASA C-MAPSS FD001 benchmark dataset.

The experimental results and residual analysis confirmed that:

1. The model effectively filters out sensor noise and captures long-term degradation trends.
2. The dynamic learning rate scheduling ensures convergence to a stable minimum, avoiding overfitting.
3. The framework serves as a reliable health monitoring tool, capable of distinguishing between safe operation zones and critical failure onsets, as demonstrated in the case studies.

Future Work: Future research will focus on extending this methodology to more complex operating conditions (e.g., FD002 and FD004 datasets) with multiple fault modes. Additionally, we aim to explore Attention Mechanisms and Transformer-based architectures to further improve interpretability by identifying the specific temporal moments that contribute most to the failure prediction.

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